

TASK 1

I calculated the adversity percentage for each client at horizons $\tau=5, 10, 15, 20, 25, 30$. The adversity percentage represents how often a client's trades become adverse for the Liquidity Provider after the given time horizon.

The main trend is that adversity generally increases as the horizon becomes longer. This means that the longer the LP holds the position after trading with the client, the higher the chance that the trade moves against the LP.

Client A has the lowest and most stable adversity profile, staying around 40–42% across all horizons. This suggests Client A is comparatively less toxic or less adverse for the LP.

Clients B and C show a moderate increase in adversity as the horizon increases. Their trades become slightly more adverse over time, but the increase is still gradual compared to the later clients.

Clients D, E, and F show a much stronger increase in adversity percentages. In particular, Client F has the highest adversity, increasing from around 47% at $\tau=5$ to almost 62% at $\tau=30$. This indicates that Client F is the most adverse client for the LP, especially at longer horizons.

Overall, the client profiles show a clear ordering from least adverse to most adverse: Client A is the safest, followed by B and C, while D, E, and especially F are more costly from the LP perspective.